Masih Daneshfar

Master's Candidate in Nanoscale Robotics & Automation · Software Engineer @ SmarAct · M.Sc. Engineering of Socio-Technical Systems

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ABOUT

Master's student in Systems Engineering with a focus on embedded and high-precision systems. Currently a Software Engineer at SmarAct, working on a Master's thesis dedicated to building an automation framework for nanoscale movements.

Solid foundation in micro-robotics and microscopy principles (SEM, SPM, FIB/EBiD), with a focus on bridging robust software solutions and complex physical hardware. Always looking for new challenges and ways to keep learning.

RESEARCH INTERESTS

Precision control engineering and systems integration for the nanoscale. My work focuses on bridging software architecture with physical hardware to manipulate and characterize nanomaterials. I am particularly interested in closed-loop piezo-control, machine-vision integration, and applying systems engineering principles to interdisciplinary nano- and quantum research.

MASTER'S THESIS

Automated Robotic Framework for Nanowire Pick-and-Place Integration (working title)

Oct 2026 — Summer 2027

Microrobotics and Control Engineering Department (AMiR Lab) / Carl von Ossietzky University of Oldenburg

Ex-situ nanowire pick-and-place under an optical microscope, using a 6-DoF piezo-positioning setup. Porting and extending an existing Python-based teleoperation framework to a new SDK environment, then building toward scripted and machine-vision-guided automation.

6-DoF Piezo-Positioning · Goniometer Alignment · Python SDK Integration · Teleoperation Frameworks · Machine-Vision Automation

EXPERIENCE

Student Software Engineer

Oct 2025 — Present

SmarAct Group · Part-time

Oldenburg, Lower Saxony, Germany · Hybrid

Engineering desktop applications and UI architecture for precision positioning systems by implementing C++/Qt/QML components.

C++ · Qt/QML · Python · CMake · Git

Student Research Assistant

Jul 2025 — Nov 2025

Carl von Ossietzky University of Oldenburg · Part-time

Oldenburg, Lower Saxony, Germany

Investigated deterministic network infrastructure, host-switch topologies, and simulation modeling.

OMNeT++ · Time-Sensitive Networking (TSN) · Simulation

VOLUNTEERING

RISE — Researchers for Inclusion, Support & Exchange

Jul 2025 — Present

Quantum Valley Lower Saxony

Supported the planning and website for RISE, a new department at QVLS built around inclusion, support, and exchange within the quantum community.

EDUCATION

Carl von Ossietzky University of Oldenburg

Oct 2024 — Present

M.Sc. Engineering of Socio-Technical Systems (Specialization: Systems Engineering)

SKILLS & METHODOLOGIES

Software & Embedded Systems
C++ · Qt / QML · Python · Machine-Vision · ROS2 · Git

Nanotechnology & Characterization Principles
Scanning Electron Microscopy (SEM) · Scanning Probe Microscopy (AFM / STM) · FIB / EBiD · Thin-Film Sputtering

Thesis & Lab Applications
6-DoF Piezo-Stage Control · Micro-Robotic Teleoperation · Optical Vision Feedback · Micro-force & Adhesion Dynamics